HackathEOn

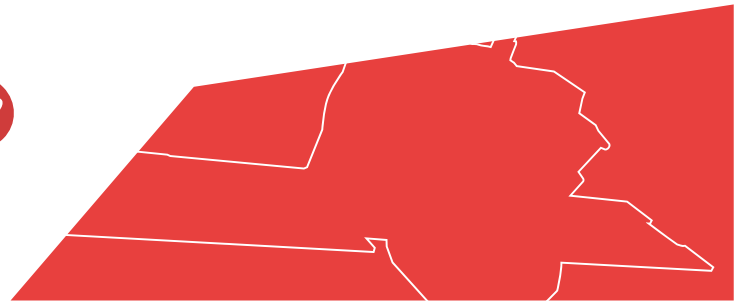
Earth Observation Center

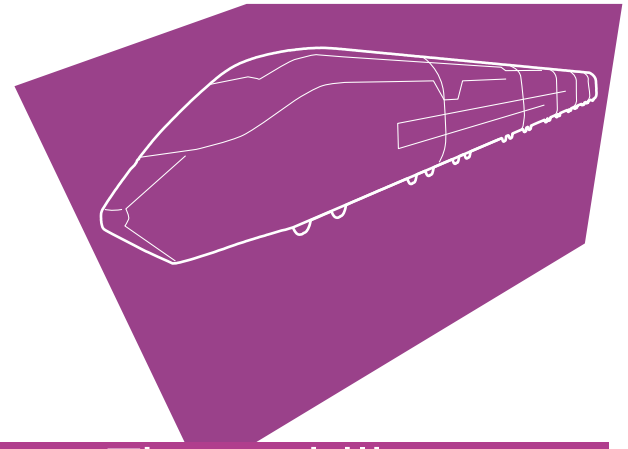
DLR



Economical
and efficient
flight

High-tech for
spaceflight – near
and far from the Earth



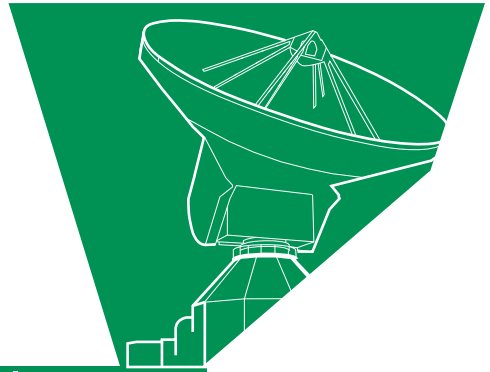


The mobility
of the future

DLR Deutsches Zentrum für Luft- und Raumfahrt

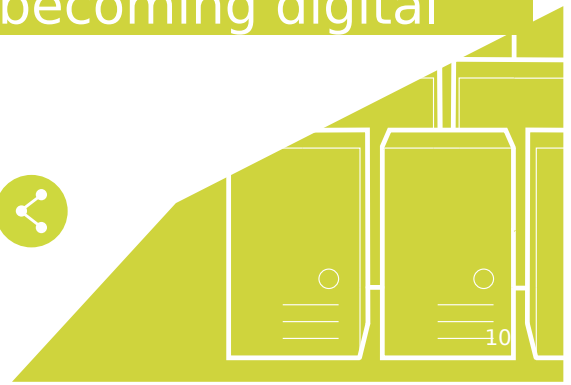
Sustainable
energy systems





Technology
for security

The world
is becoming digital



DLR in Numbers

- 30 locations across Germany
- 4 offices abroad
- 54 research institutes
- 11,000 employees



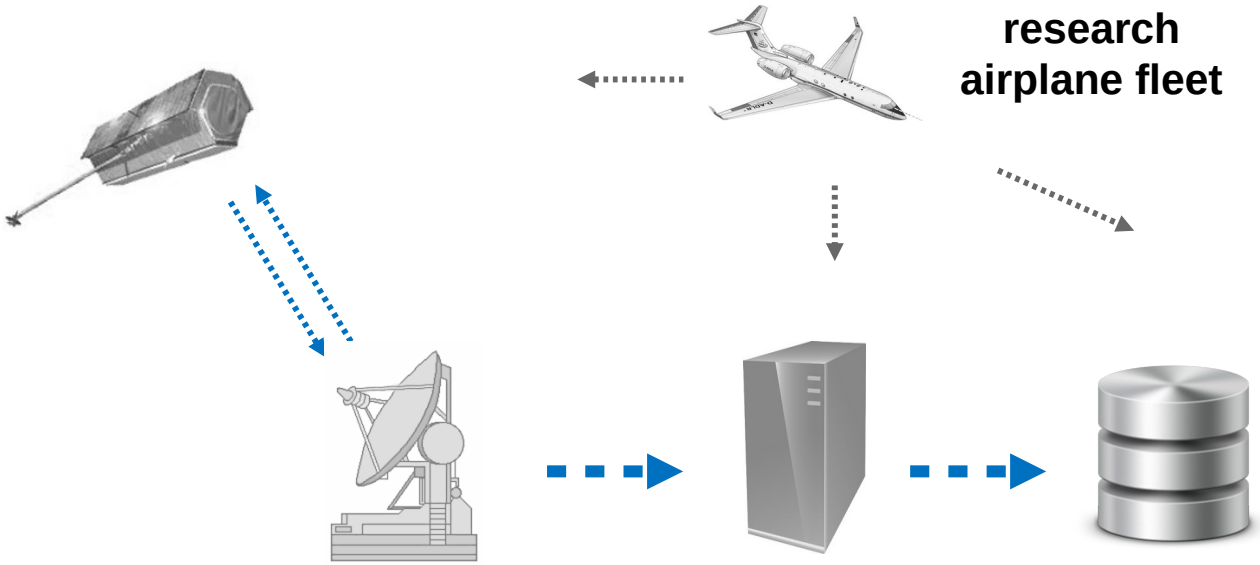
SPACE
ENERGY
GERMAN SPACE AGENCY AT DLR
AERONAUTICS
DIGITALISATION
TRANSPORT
SECURITY
DLR PROJECT MANAGEMENT AGENCY

German Remote Sensing Data Center (DFD)
Prof. Dr. Christian Geiß
(designated)

Remote Sensing Technology Institute (MF)
Prof. Dr. Michael Eineder
(acting)



Core Competence in EO



ground segment and services

sensors and missions

information retrieval

geoscientific research

data

information products

S/W and H/W

scientific results

services

scientists

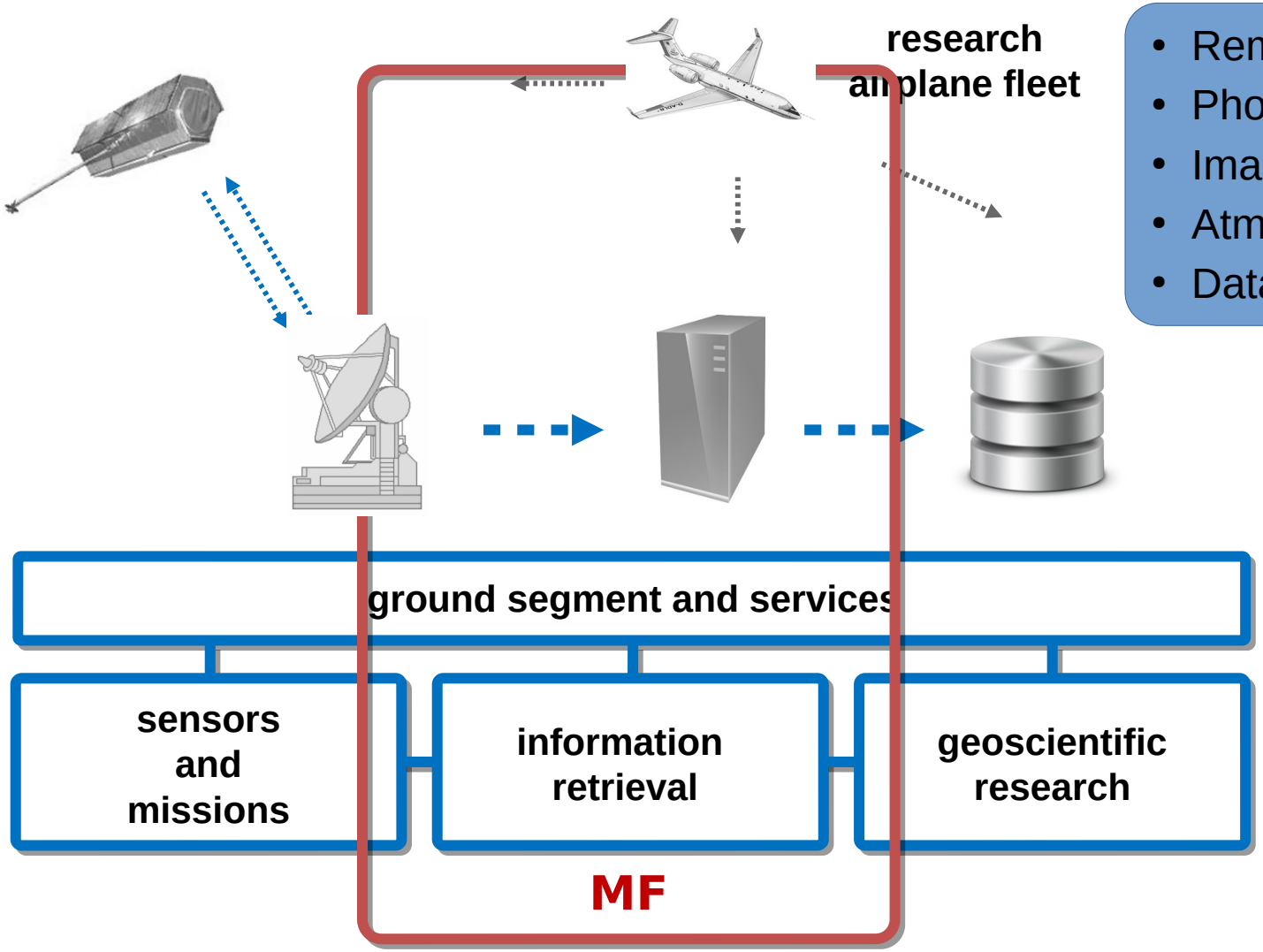
agencies

industry

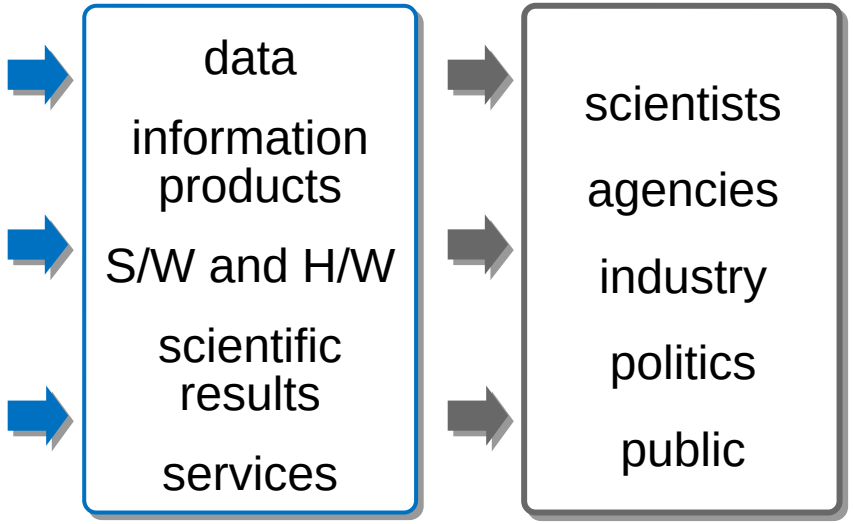
politics

public

Core Competence in EO of MF



- Remote sensing with Synthetic Aperture Radar
- Photogrammetry and Image Analysis
- Imaging Spectroscopy
- Atmospheric Spectroscopy
- Data Science and Artificial Intelligence

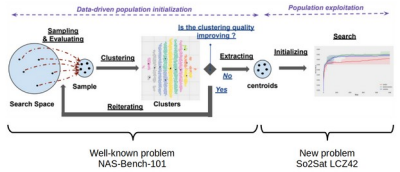


EO Data Science

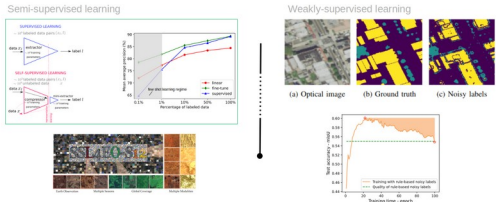
EFFICIENCY

(ENERGY & *HUMAN*) (DATA)

Data-driven neural architecture search initialization
Deploy centroids of a well-known problem to a new one

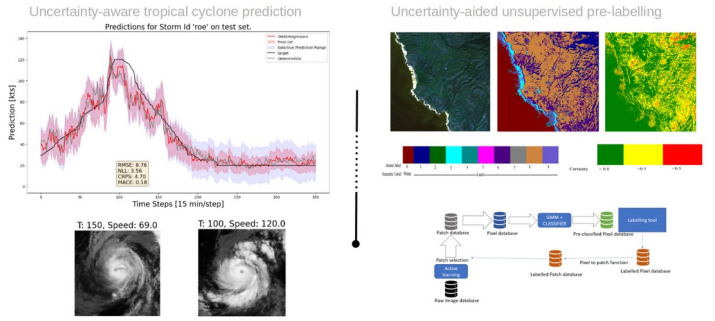


Making a good use of limited labeled data
and a huge amount of unlabeled data

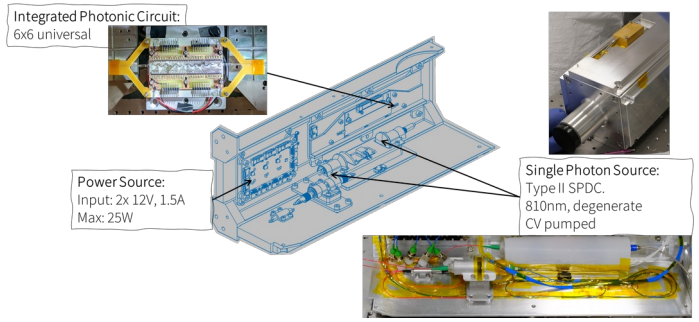


TRUSTWORTHINESS

Quantifying uncertainty on ML-based models for EO applications



QUANTUM COMPUTING & SECURITY



KNOWLEDGE TRANSFER



Traoré, Camero, and Zhu, Annals of Mathematics and Artificial Intelligence, 2022
Lehmann et al., ICLR, 2024

Wang et al., IEEE GRS, 2022.
AutoCoast Project – Karmakar et al., ATOMS, 2024

Liu et al, IEEE Big Data, 2022.
Steiner et al. ArXiv 2026

EO Data Science

EFFICIENCY

(ENERGY & *HUMAN*) (DATA)

Data-driven neural architecture search initialization

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TRUSTWORTHINESS

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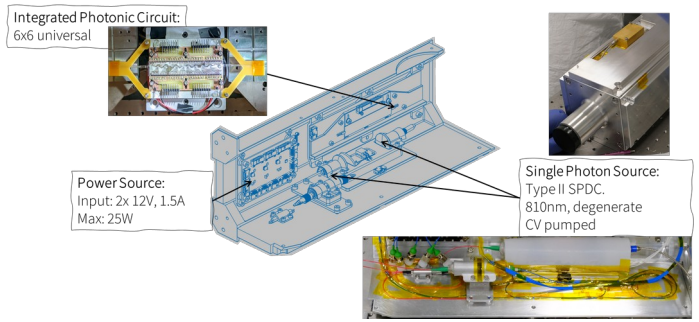
Uncertainty-aware tropical cyclone prediction

Predictions for Storm Id 'roe' on test set.

Uncertainty-aided unsupervised pre-labelling

On-board Processing

Geo-Foundation Models





Traoré, Camero, and Zhu, Annals of Mathematics and Artificial Intelligence, 2022
Lehmann et al., ICLR, 2024

Wang et al., IEEE GRS, 2022.
AutoCoast Project – Karmakar et al., ATOMS, 2024

Liu et al, IEEE Big Data, 2022.
Steiner et al. ArXiv 2026

HackathEOn

Agenda - Monday

Time	Topic
09:00 - 09:15	HackathEOn introduction
09:15 - 10:45	AI4EO at a glance
10:45 - 11:15	Challenge introduction
11:15 - 11:30	Break
11:30 - 13:00	Classification
13:00 - 14:00	Break
14:00 - 15:30	Segmentation
15:30 - 15:45	Break
15:45 - 17:15	Segmentation pt. 2
17:15 - 17:30	Day 1 wrap up

Agenda - Tuesday

Time	Topic
09:00 - 10:30	UQ
10:30 - 11:00	Break
11:00 - 12:30	Geo-FM
12:30 - 13:30	Break
13:30 - 13:45	Team building
13:45 - 17:15	Group work + Q&A
17:15 - 17:30	Day 2 wrap up

Agenda - Wednesday

Time	Topic
09:00 - 09:15	Welcome
09:15 - 10:30	Group work
10:30 - 11:00	Prepare presentations
11:00 - 11:45	Presentations
11:45 - 12:00	Break
12:00 - 12:30	Awards and closing remarks



Dr. Anja Rösel



Dr. Jakob Fischer

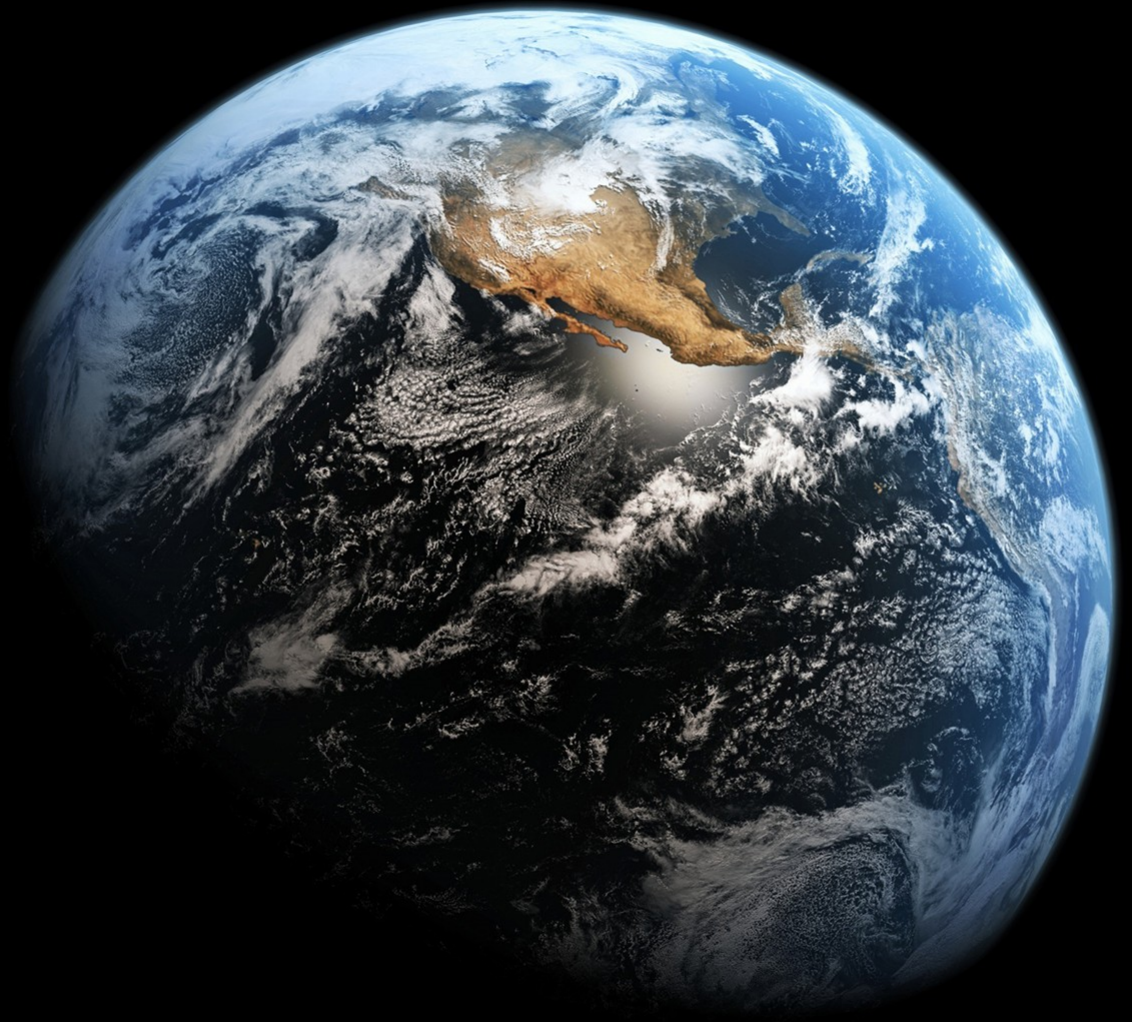


Dr. Vytautas Jancauskas



Dr. Andrés Camero

For Earth into Space



eoc

DLR